Lid Muscle Abnormalities

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Lecture Acknowledgement and Thanks:

Professor William C. Hefner

Pacific University



SOAP Format

- Subjective
- Objective
- Assessment
- Plan

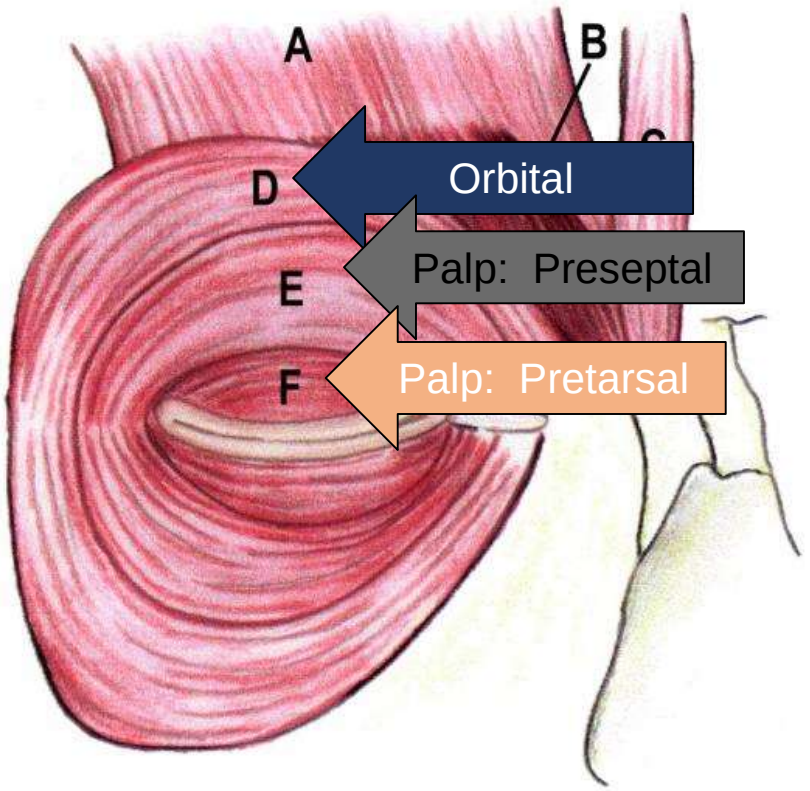


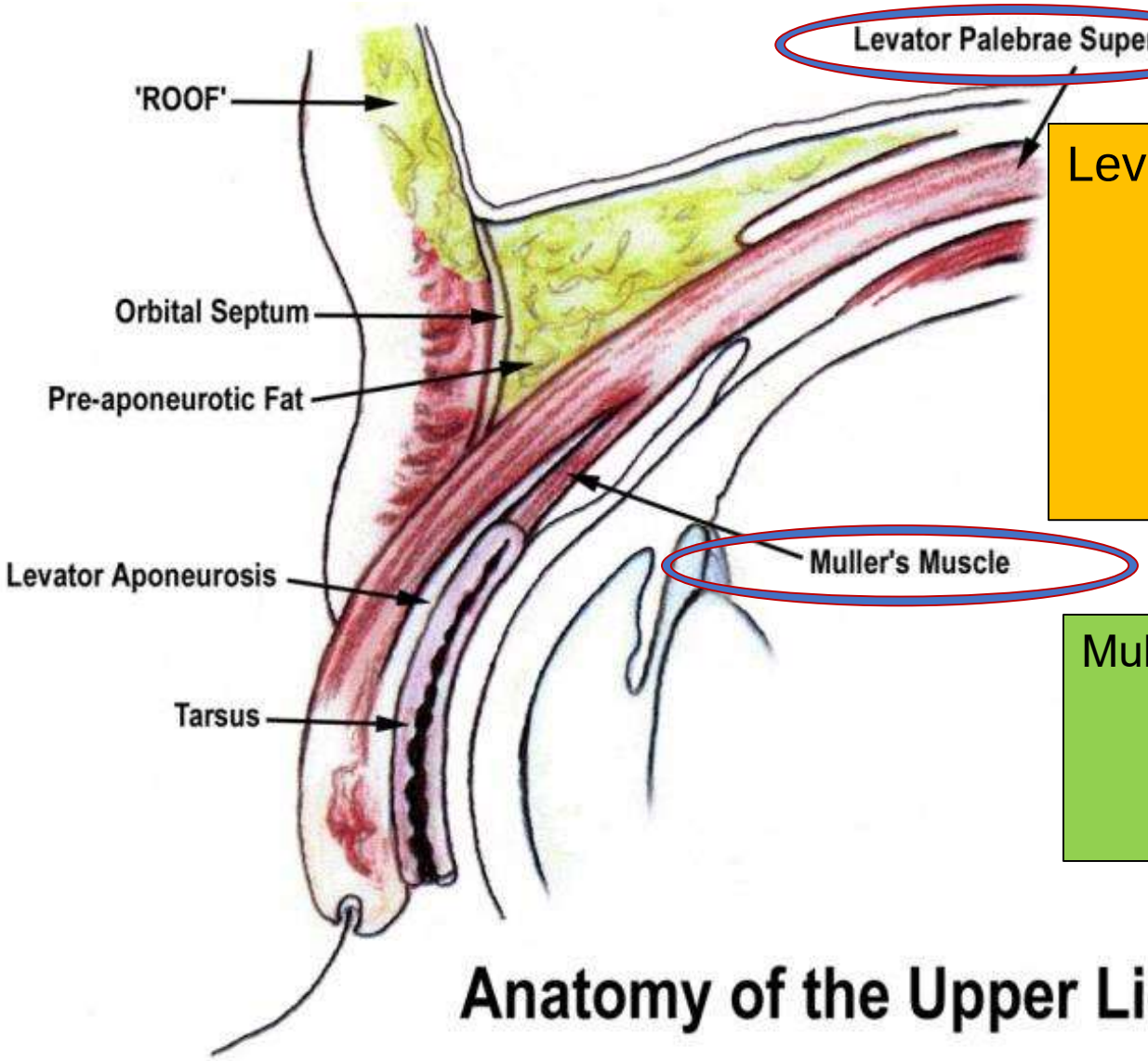
Eyelid basics

- Interpalpaebral Fissure Width (Height) (aka PFH):
  - Young adults 10-11 mm
  - Older adults 8-10 mm
- Upper lid margin 1-2 mm BELOW s. limbus
- Tarsal plates are ~25 mm horizontally
  - Meibomian glands housed within
    - ~25 in upper lid
    - ~20 in lower lid

Eyelid basics- the muscles

- Orbicularis oculi
  - Orbital portion
  - Palpaebral portion
    - Pre-Septal
    - Pre-Tarsal
    - Ciliary (aka Riolan Muscle)
  - Lacrimal portion
- Levator Palpebrae
- Muller's





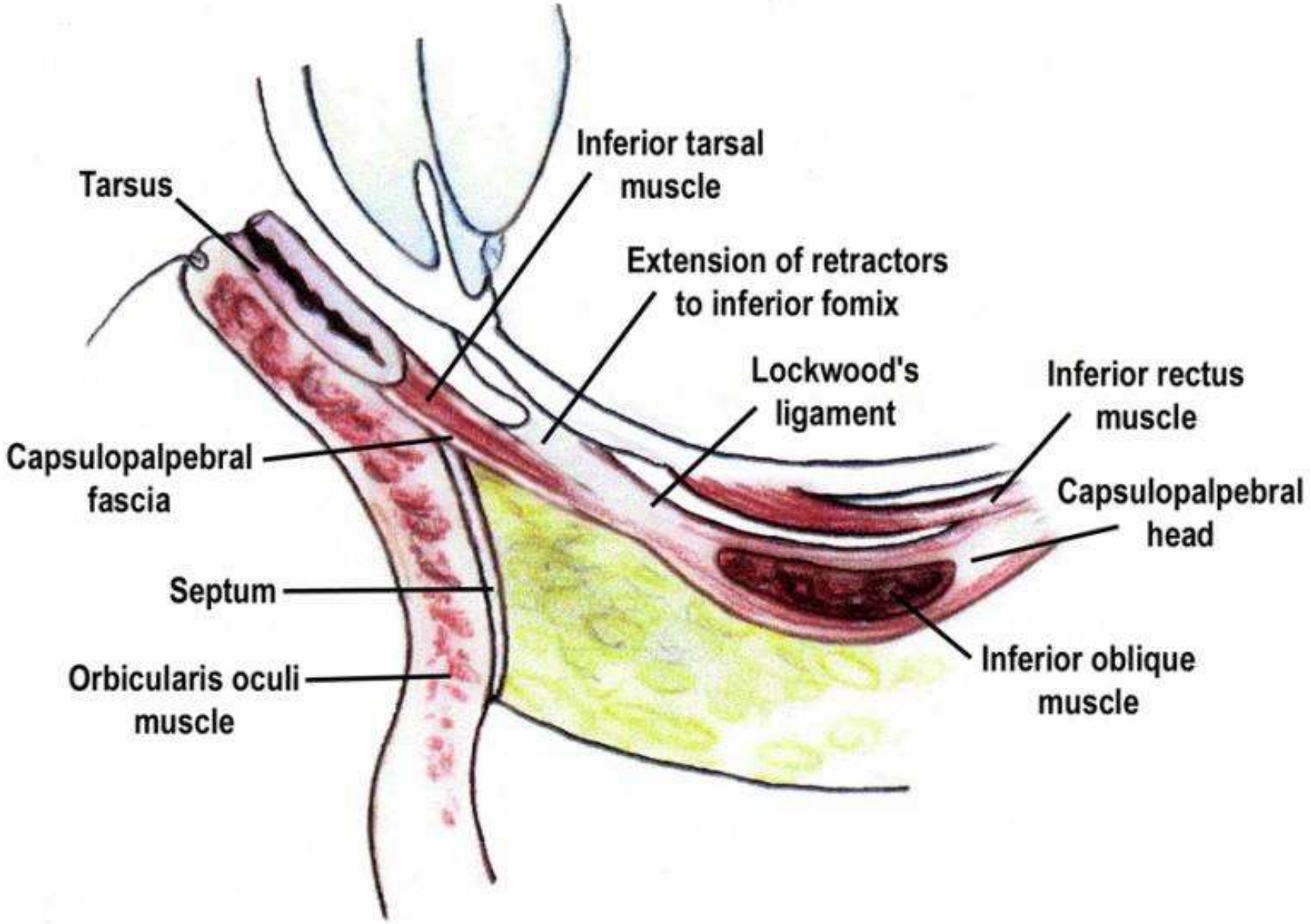
Levator:

- Skeletal muscle
- Opening the eye
- Innervation is superior branch of CN III
- Raises lid ~15 mm

Muller's:

- Smooth muscle
- Innervation is CN III
- Raises lid ~1-2 mm

http://emedicine.medscape.com/article/834932-overview#showall



Anatomy of the Lower Lid

http://emedicine.medscape.com/article/834932-overview#showall

Lid position determined by:

1. Tone of Orbicularis Oculi Muscle

2. Tone of Levator Palpaebrae Muscle



s this ma



Normal?

Abnormal?



http://webeye.ophth.uiowa.edu/eyeforum/tutorials/ptosis/2a-LRG-CN3.jpg

Roles of the lids

- Protection
- Tears:
  - Production
  - Distribution
  - Removal
- Cosmesis

What do you think the patient is experiencing? (S)



https://www.seattleface.com/service/face-and-neck-surgery/ptosis-repair

What is the problem here? (O)



https://www.seattleface.com/service/face-and-neck-surgery/ptosis-repair

What do you think caused the problem? (A)



https://www.seattleface.com/service/face-and-neck-surgery/ptosis-repair

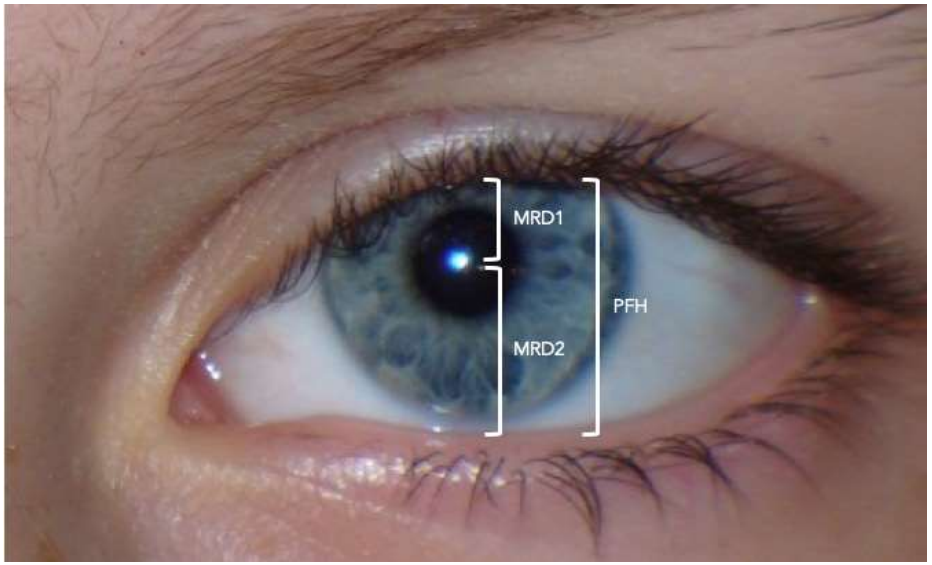
How do you want to fix this? (P)



https://www.seattleface.com/service/face-and-neck-surgery/ptosis-repair

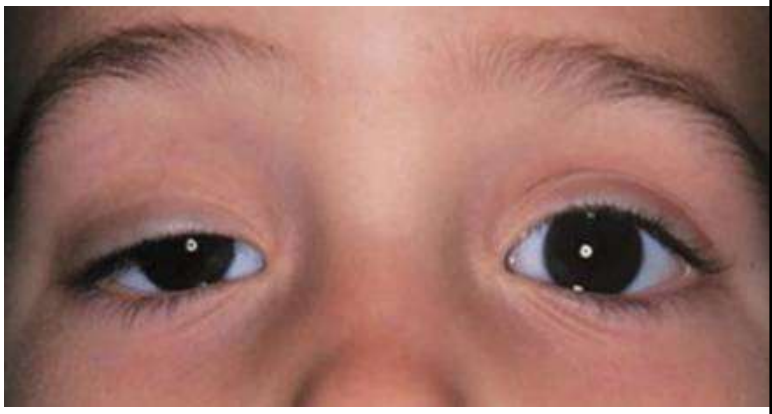
Ptosis: Abnormally low position (drooping) of upper eyelid





- Also called blepharoptosis
- Types
  - Congenital
  - Acquired

Congenital Ptosis



- Present at birth
- 90% ptosis cases
- 70% unilateral
- 30% have strabismus
- 12% have anisometropia >1.00D
- 20% have amblyopia
  - Typically deprivational

Congenital Ptosis

Causes

1. Myogenic (Irregularity of levator muscle)
2. Neurogenic (Abnormality of CN III superior division)
3. Anomalous Innervation (Marcus Gunn Jaw Winking)






Congenital Ptosis

- Myogenic (Irregularity of the levator muscle)
  - Majority of congenital ptosis cases
  - Present at birth, stable throughout life
  - Upper eyelid crease normally ABSENT

Lid position in downgaze should be noted

  - In myogenic-caused congenital ptosis, ptotic lid appears higher in downgaze
  - In acquired ptosis, the ptotic lid is equal to the non-ptotic lid



Congenital Ptosis

- Neurogenic (Abnormality of CN III superior branch)
  - Disrupted innervation to Muller's muscle
  - Most common occurrence is Congenital Horner's
    - Ptosis, miosis, anhidrosis and often iris heterochromia
    - Look for Harlequin Sign (flush of unaffected side of face)



Morrison DA, Bibby K, Woodruff G. The "harlequin sign and congenital Horner's syndrome. *J Neurol Neurosurg Psychiatry. 1997;*62:626-628

Congenital Ptosis

- Anomalous Innervation
  - Marcus Gunn “jaw-winking” syndrome
    - Chewing/moving jaw to opposite side → lid elevation
    - Misdirection of either:
      - Efferent motor innervation
      - Afferent proprioceptive fibers of CN III and V



https://collections.lib.utah.edu/details?id=188578&q=eyelid+disorder

Acquired Ptosis

- Results from:
  - Affliction of nerve supply to lid musculature
  - Diseases of lid muscles
  - Mechanical interferences in elevation

Acquired Ptosis



- Causes:
  - Neurogenic
  - Myogenic
  - Aponeurotic
    - Where levator aponeurosis has undergone structural changes which reduces its ability to lift eyelid
  - Mechanical

Acquired Ptosis

- Neurogenic
  - Horner's syndrome
    - Ipsilateral findings of:
      - mild ptosis
      - miosis
      - anhidrosis



https://www.semanticscholar.org/paper/Orbital-rhabdomyosarcoma.-0ffd0197821f76

https://openi.nlm.nih.gov/detailedresult.php?img=PMC3086325_SHORTS-10-10101&req=4

Acquired Ptosis



https://www.semanticscholar.org/paper/Orbital-rhabdomyosarcoma.-Karcioglu-Hadjistilianou/b3f8f8b31ff83e2f92b9e5f2ad0ffd0197821f76

Acquired Ptosis

- Neurogenic
  - Deficient innervation of CN III to levator
  - Most common associated conditions:
    - Diabetes mellitus (DM)
      - “Down and out” affected eye (CN III)
    - Horner’s Syndrome
      - “Ptosis, miosis, anhidrosis” affected side
    - Aneurysms of internal carotid artery

Acquired Ptosis (Neurogenic)



Defective Adduction



Defective Elevation



Defective Depression



Normal Abduction

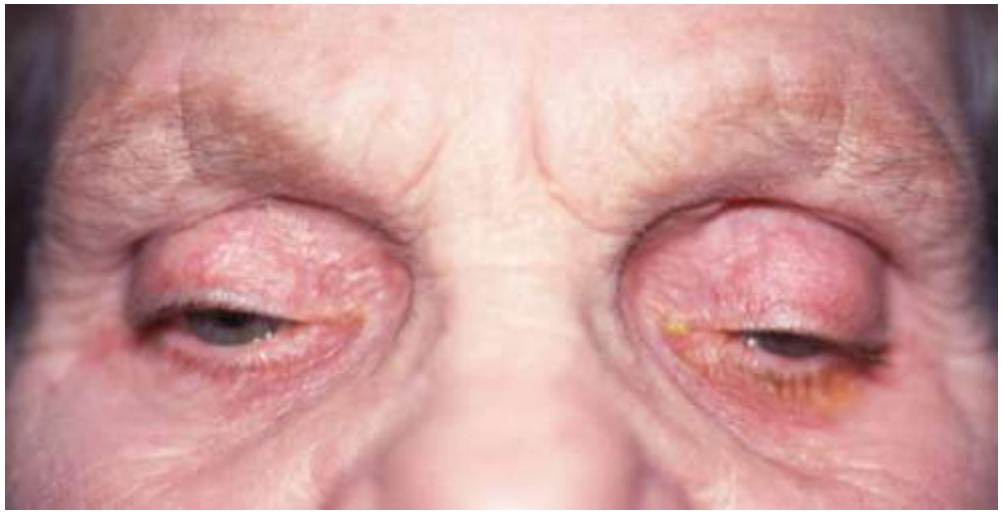
Acquired Ptosis

- Myogenic
  - Myasthenia Gravis
    - Abnormality at neuromuscular junction
    - Fatigability of voluntary muscle (Tensilon, cold pack test)
    - 96% have ptosis: bilateral but asymmetrical
    - Pt may have history of fluctuating ptosis & strabismus



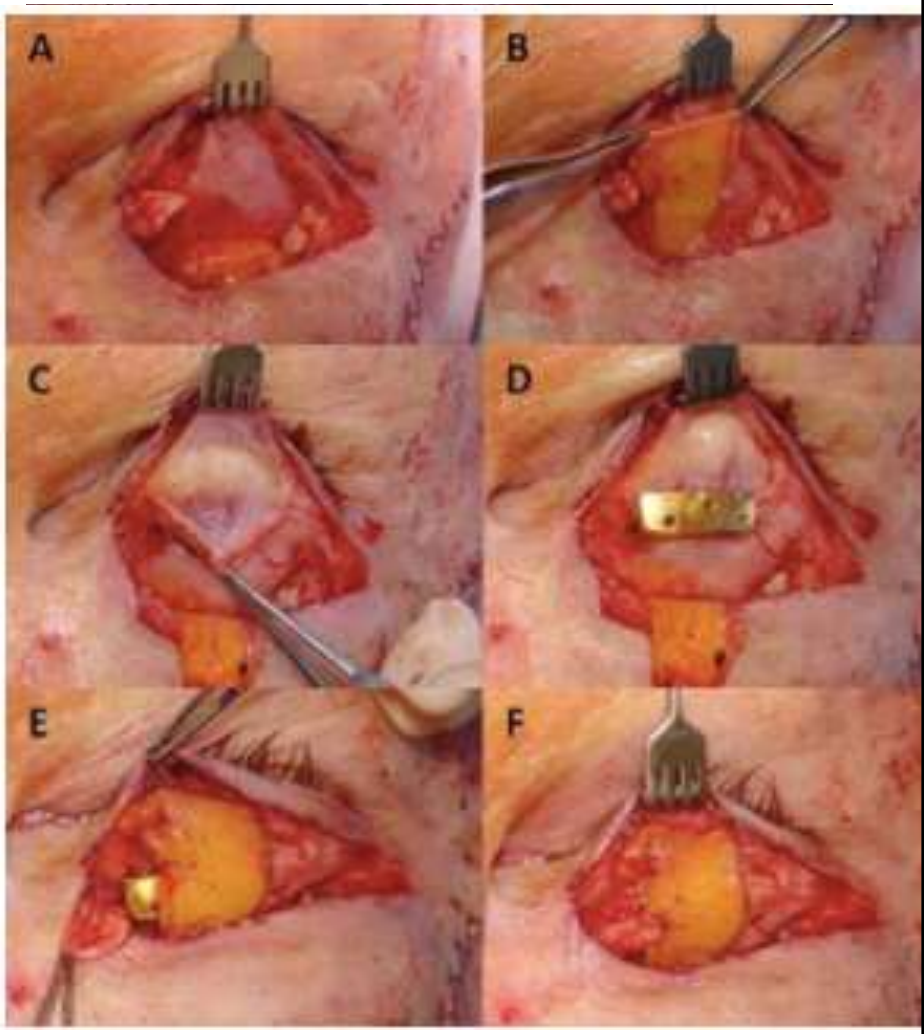
Acquired Ptosis

- Aponeurotic Dehiscence
  - Loss of levator tone and/or degeneration of levator attachments
    - Usually bilateral
    - Most common acquired ptosis



Acquired Ptosis

- Mechanical
  - Excessive weight
    - Edema
    - Tumors
    - Eyelid plates
  - Traumatic
    - Lacerations effecting nerves or muscles



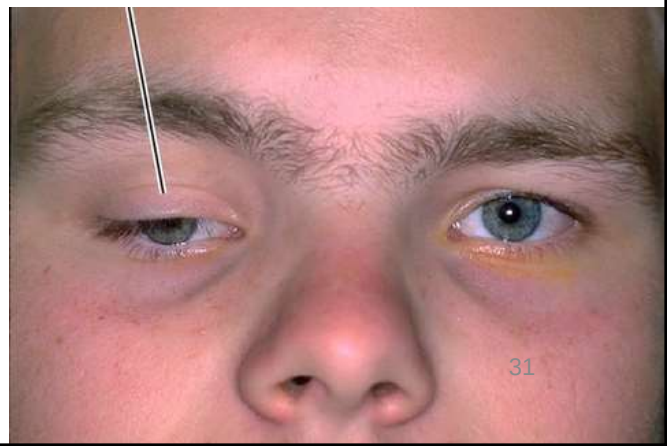
Ptosis vs. Pseudoptosis

- Upper lid looks low, lid retraction is normal
- Pseudo-ptosis conditions:
  - Dermatochalasis
  - Enophthalmos
  - Hypotropia
  - Prosthetic eye









Ptosis

- Clinical Evaluation
  - HX
    - Onset: When first noticed?
    - FMHx: Anyone in family with ptosis?
    - Diplopia: Double vision?
    - Variability: Does it change throughout day?
    - Systemic symptoms/conditions
    - Trauma history
    - Allergic or drug reaction history – may cause edema

Ptosis

- Clinical Evaluation
  - Objective
    - Ptosis amount
    - Levator function
    - Pupils

Ptosis

- Clinical Evaluation (continued)
  - Ptosis amount
    - Measure both palpebral fissures and note the difference

Severity	Difference between fissures
Mild	1 – 2mm
Moderate	2 – 3 mm
Severe	≥ 4 mm

Ptosis

- Clinical Evaluation (continued)
  - Levator Function

Place ruler with 1mm marking at upper lid margin in downgaze



Normal levator function is 15 mm

Have patient look up. Difference between the downgaze & upgaze = function



Ptosis



- Clinical Evaluation (continued)
  - If congenital ptosis obscures any part of pediatric patient's visual field, surgery must be performed to correct problem early in life
    - Otherwise, permanent vision loss may occur as result of amblyopia:
      - Occlusion (deprivational) amblyopia
      - Astigmatism from compression by droopy eyelid

Ptosis

Differentials:

- Chronic Progressive External Ophthalmoplegia
- Myasthenia Gravis
- Papilledema
- Acquired myogenic ptosis
- Aponeurotic ptosis
- Neurogenic ptosis
- Mechanical ptosis
- Traumatic ptosis
- Pseudoptosis
- Blepharochalasis
- Myotonic dystrophy
- Kearns-Sayre syndrome
- Acquired blepharoptosis

Ptosis

- Non-Surgical Treatment
  - Mild Cases
    - Observation if no signs of amblyopia, strabismus, and abnormal head posture

Ptosis

- Non-Surgical Treatment (continued)
  - Mild cases



www.pediatriceyemd.com

  - Follow-up every 3-4 months for signs of amblyopia due to congenital ptosis

Ptosis

- Non-Surgical Treatment (continued)
  - Head posture should be carefully examined
    - If patient acquires a chin-up posture due to worsening of ptosis, surgery may be indicated
  - Patient should be checked for astigmatism due to compression by droopy eyelid

Ptosis

- Surgical Treatment
  - Functional reasons
  - Physical reasons
  - Psychological reasons



www.tedmontgomery.com

Ptosis

- Pre-Surgical Treatment should include:
  - VA's
  - Visual fields (threshold VF)
    - Untaped
    - Taped
    - Must demonstrate VF impingement

Ptosis

- Surgical Treatment
  - Levator shortening
    - Must have some levator function
  - Frontalis suspension procedure
    - Levator function must be poor
    - The lids are linked to frontalis movement



www.youreyesbydrmary.com





https://www.seattleface.com/service/face-and-neck-surgery/ptosis-repair

Ptosis summary

- Lid droop
- Congenital
- Acquired
  - Multiple types
- Careful history and follow-up
- Treatment ranges from reassurance to surgery

Blinking

- Average rate 15-20 blinks/minute
- One blink takes about 0.25 seconds
- Disease may alter rates (ex. Parkinson's, Hyperthyroid)
- Partial blinks
- Aggravated by keratoconjunctivitis sicca (dry eye)
  - Remember contact lens wearers

Blepharoclonus

- Increased frequency of blinking
- Increased closure phase
- Caused by
  - Irritation
  - Inflammation
- Most frequently seen in children

Myokymia

- Orbicularis fasciculation
- “An annoying twitch”

- Etiology:
  - Stress, tension, lack of sleep, caffeine, etc.

Myokymia

- Therapy:
  - Remove cause
  - Botulinum toxin type A (Botox)
    - Injected into affected area
    - 2.5-5U IM q3mo

Myokymia



- Treatment (continued)
  - Antihistamines
    - Antazoline (Vasocon A) q4h
    - Pheniramine (Opcon A, Naphcon A, Visine A) q4h
    - Promethazine (Phenergan) 12.5-25mg PO qd-tid
  - Quinine 200-300mg PO qd-bid
    - Tonic Water!!!!!

Blepharospasm



- Definition: Any abnormal tic or twitch of eyelid
- Bilateral involuntary orbicularis contraction: results in eye closure
- Spasmodic
- Idiopathic

Blepharospasm



- Patients over 50
- F>M
- There are at least 50,000 cases of blepharospasm in United States
  - Up to 2000 new cases diagnosed annually
  - Prevalence approximately 5 in 100,000

Blepharospasm

- Subjective
  - At one end of spectrum - essential blepharospasm - manifested by simple increased blink rate and intermittent eyelid spasms
  - On the other end of the spectrum - blepharospasm is disabling condition with ocular pain and functional blindness
  - Blepharospasm is commonly associated with dystonic movements of other facial muscles

Blepharospasm

- Subjective (continued)
  - Changes associated with long-standing blepharospasm include:
    - Eyelid and brow ptosis
    - Dermatochalasis
    - Entropion
    - Canthal tendon abnormalities

Blepharospasm

- Early symptoms of blepharospasm include:
  - Increased blink rate (77%)
  - Eyelid spasms (66%)
  - Eye irritation (55%)
  - Midfacial or lower facial spasm (59%)
  - Brow spasm (24%)
  - Eyelid tic (22%)

Blepharospasm

- Methods to relieve blepharospasm include:
  - Sleep (75%)
  - Relaxation (55%)
  - Inferior gaze (27%)
  - Artificial tears (24%)
  - Traction on eyelids (22%)
  - Talking (22%)
  - Singing (20%)
  - Humming (19%)

Blepharospasm

- Comorbid diagnoses include:
  - Dry eyes (49%)
  - Other neurologic disease (8%)
  - Video: http://www.kaltura.com/index.php/extwidget/preview/partner_id/797802/uiconf_id/27472092/entry_id/0_ts42qgoe/embed/auto?

Blepharospasm

- Differential Diagnoses:
  - Secondary to:
    - Foreign body (FB)
    - Trichiasis
    - Keratitis
  - Meige's syndrome
  - Hemifacial spasms
  - Facial Myokymia
  - Eyelid Myokymia
  - Facial Tic



Blepharospasm

- Therapy
  - Reassurance!
  - Oral meds are not too effective
  - Botulinum toxin type A (Botox)
    - Injected along upper and lower eyelid
    - 20-25U per eye IM q3mo
    - Most require re-injection in 3-4 months

Blepharospasm

- Therapy (continued)
  - Surgical
    - Last choice in therapy
    - Involves removal of orbicularis
    - Reserved for those unresponsive to botulinum toxin

Blepharospasm DDx

Differential Diagnosis	Gender	Age	Laterality	Voluntary Control	Clinical Features
Essential Blepharospasm	**F>M**	**>50**	**Bilateral**	**No**	**Isolated spasms of orbicularis**
Meige's Syndrome	**F>M**	**>50**	**Bilateral**	**No**	**Blepharospasms and midfacial spasms**
Hemifacial Spasm	**F>M**	**>45**	**Unilateral**	**No**	**Spasms along the distribution of CNVII**
Facial Myokymia	**F=M**	**Any**	**Unilateral**	**No**	**Rapid muscle flickering**
Eyelid Myokymia	**F=M**	**Any**	**Unilateral**	**No**	**Fasciculations of the orbicularis**
Facial Tic	**F=M**	**Youth**	**Bilateral or Unilateral**	**Yes**	**Repetitive movements**

Video Resource

Link to multiple videos of ocular and other neurological problems:

http://content.lib.utah.edu/cdm4/browse.php?CISOROOT=/ehsl-shw

THANK YOU!